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When an AI Says It Is False: User Responses to Misinformation Flagging by Automated vs. Human Fact-Checkers

Mengqi Liao^a, Sian Lee^b, Annie Dooley^c, Aiping Xiong^d, and S. Shyam Sundar^{e,f}

^aDepartment of Advertising & Public Relations, Grady College of Journalism & Mass Communication, University of Georgia, Athens, GA, USA; ^bDepartment of Scholar Support and Data Services, University of Mississippi, Oxford, USA; ^cSchool of Communication, College of Arts and Sciences, The Ohio State University, Columbus, USA; ^dCollege of Information Sciences and Technology, The Pennsylvania State University, PA, USA; ^eDonald P. Bellisario College of Communications, The Pennsylvania State University, PA, USA; ^fDepartment of Immersive Media Engineering, Sungkyunkwan University, Seoul, Republic of Korea

ABSTRACT

To combat misinformation at scale, automated fact-checkers are being deployed, but we do not know if lay users trust them. Are AI fact-checkers trusted more than human fact-checkers because of their accuracy in identifying tell-tale features of fake news? Or are they trusted less because they are seen as lacking the subjectivity necessary for corroborating evidence? A pre-registered 2 (Fact-checking source: Human vs. AI) × 3 (Fact-checking approach: Evidence-based vs. Feature-based vs. Black-box) between-subjects experiment among 291 US adults recruited from Cloud Research revealed that users' trust was predicted by the extent to which the interface triggered the positive machine heuristic (the algorithm is more objective and precise than human) and the negative machine heuristic (the algorithm lacks human subjective judgment). The latter was more likely when the system used an evidence-based determination of misinformation, which was better understood by users than a feature-based approach. Theoretical and practical implications for individuals' trust of automated fact-checkers are discussed.

Online misinformation has become rampant in a variety of domains. Misinformation is any false or inaccurate information that is deliberately created and is intentionally or unintentionally propagated (Wu et al., 2019), posing significant challenges to media trust, decision-making, social cohesion, public health, democracy, and cybersecurity (Saeidnia et al., 2025).

To reduce user engagement with online misinformation, platforms commonly employ debunking mechanisms such as fact-checking and issuing

CONTACT Mengqi Liao  mengqi.liao@uga.edu  Department of Advertising & Public Relations, Grady College of Journalism & Mass Communication, University of Georgia, Journalism Building, Room 223-A, GA 30602, USA
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warnings or community notes for users. Social media companies often rely on third-party fact-checkers to evaluate suspicious claims or encourage expert users to provide community notes regarding the veracity of online posts. These fact-checking results are also accompanied by explanations regarding why specific content was flagged as false, such as who (the expertise of the fact-checker) made the decision and how particular wording or claims appeared suspicious (Kotonya & Toni, 2020). A recent meta-analysis of 21 experiments found that fact-check labels significantly reduced both belief in and sharing of false news, even among users who are most distrustful of fact-checkers (Martel & Rand, 2024). However, by the time human fact-checkers or community members verify the information, misinformation has often already spread across the internet, causing harm (Walter & Tukachinsky, 2020). Therefore, increasingly, in addition to relying solely on human efforts, researchers and platforms have explored the development of machine learning algorithms to detect misinformation in a more automatic, timely, and efficient manner (Reis et al., 2019). Machine learning models are particularly well suited for this task, as they can process large volumes of posts at high speed, thus ensuring scalability of fact-checking in real time (Chuai et al., 2024).

However, users may not be ready to trust automated fact-checkers just yet. Research on trust in machine classification of misinformation is equivocal (Jia et al., 2022; Seo et al., 2019; Yaqub et al., 2020; Zhang et al., 2021). One possible explanation is the difference in approaches between automated fact-checkers and humans. While human fact-checkers predominantly use an evidence-based approach that relies on the individual's corroboration of past evidence from reliable sources, automated fact-checkers usually rely on a feature-oriented approach that makes the classification based on specific manifest features. Furthermore, the mere identification of an AI as the fact-checker, as opposed to a human expert, could affect user perceptions of the system and dictate their acceptance of the result of the fact-check. In other words, both the source of fact-checking (i.e., human vs. AI) and the approach employed by the system can have significant psychological effects on users' trust toward fact-checking systems. It remains an open question whether users' distrust (or trust) toward AI in the fact-checking context is due to certain preconceptions they have toward AI or due to the approach that machines typically adopt. We examine both possibilities in this pre-registered study by grounding our work in theory and research described in the next section.

Literature Review

Despite increasing calls from researchers and platforms to explore machine learning algorithms to detect misinformation in a more automatic, timely, scalable, and efficient manner (Chuai et al., 2024; Reis et al., 2019), empirical research investigating how lay users perceive AI

fact-checkers remains mixed and equivocal. For instance, one study found that participants are more likely to trust fact-checking warnings from human fact-checkers compared to those from machine learning algorithms on social media (Seo et al., 2019). Similarly, scholars also found that users are more likely to discontinue their sharing of false information when a warning message comes from fact-checkers instead of AI (Yaqub et al., 2020). On the other hand, another study revealed that algorithms are no less trusted than third-party fact-checkers for vaccine-related misinformation (Zhang et al., 2021). Such contradictory findings motivate us to investigate the underlying mechanisms of users' formation of trust or distrust toward automated fact-checkers in the fact-checking context.

In an effort to theorize the underlying psychological mechanisms of users' interaction with AI-based media, Sundar (2020) proposed a framework to examine Human-AI Interaction (HAI) from the perspective of the Theory of Interactive Media Effects (TIME). Called the HAI-TIME model, this framework proposes two routes by which AI systems can affect user experience and trust: the *action route* and the *cue route*. While the action route specifies how a system could foster users' trust by offering users various action possibilities and thereby elicit their engagement, the cue route explicates how the mere visual manifestation of AI affordances and their attributes on the interface can trigger certain cognitive heuristics, which in turn influence user perceptions of AI.

The notion of "cues" comes from over four decades of research in the area of social cognition, which has demonstrated that people are "cognitive misers" (Fiske & Taylor, 1991), and will expend only the absolute minimum mental energy for processing information. When faced with situations of information overload, as often happens online, individuals are likely to rely on simple cues in the context to make quick judgments in an efficient manner rather than think through every aspect effortfully (Eppler & Mengis, 2004). Such contextual cues can be easily found on the interface of any system, including AI systems dedicated to fact-checking.

In the next section, we will discuss how the source of fact-checking (source cue) as well as the approach adopted for it (approach cue) might influence users' trust and perceptions of the system by triggering distinct cognitive heuristics. Understanding these psychological mechanisms is important, as cognitive heuristics may predict how users evaluate the credibility of explanations the system provides regarding why it judged the news to be false (Sundar, 2008), the degree to which they trust human or AI fact-checkers (Molina & Sundar, 2022), whether they perceive classification results as objective and free from partisan bias (Martel & Rand, 2024), and their tendency to rely on fact-checking systems to detect misinformation in the future (Molina & Sundar, 2022), all of which can help users combat the spread of misinformation online.

The Effects of Source Cues

One possible reason behind the conflicting results could be that AI is perceived by users to have both positive and negative attributes compared to human experts, and triggers different heuristics that might cancel each other out. When making decisions online, some users might apply the rule of thumb that machines are more objective, ideologically unbiased, error-free, and hence are more reliable than humans, which further increases users' trust toward machines (Sundar, 2008), namely the *positive machine heuristic*. This heuristic is very readily accessible in users' minds, and helps enhance users' trust toward the systems in various contexts such as online uncivil content moderation (Wang et al., 2021)

At the same time, users might also hold preconceptions relating to weaknesses of machines, such as being cold, emotionless and rigid compared to humans, namely the *negative machine heuristic* (Sundar, 2020). The negative machine heuristic might be especially easily triggered when machines are performing tasks that require human skills (e.g., subjective judgment and emotional capability), compared to those that require more mechanical skills (e.g., processing quantitative data for objective measures).

For instance, scholars have revealed that individuals tend to trust machines over humans for mechanical tasks, while they trust humans over machines for subjective or non-mechanical tasks, (Lee, 2018). Misinformation classification is likely to come under the latter category because it involves making subjective judgments about human language. This goes beyond online users. Researchers who strive to develop algorithms to classify online content also express their concerns regarding the possibility that machines might not have the ability to detect nuances that exist in human language (Gollatz et al., 2018), and human emotions (Lee, 2018). When users believe that AI lacks the required abilities to detect misinformation, they might also be less likely to rely on or feel dependent upon an AI system's help to detect misinformation online in the future.

A study testing the role of both the negative and positive machine heuristics in the context of content moderation (suicidal content and hate speech) found that both the positive and negative stereotypes about machines could be evoked when AI is the source, each influencing users' trust and evaluation toward the system in opposite directions (Molina & Sundar, 2022). In addition to trust and reliance, the triggering of both heuristics can produce other perceptual outcomes related to the fact-checking process. For example, users were more likely to doubt the argument quality of the hard news (i.e., news stories about political decisions, natural disasters, crime, economic reports, and major announcements) that is being fact-checked by AI fact-checkers compared to when human fact-checkers performed the task (Kim & Lee, 2025). However, this pattern did not hold for soft news (stories about

entertainment, arts, celebrities, and other culture-related subjects), which, according to the authors, may be because soft news requires more subjective human judgments, whereas hard news is often perceived as more objective and bias-free.

This finding aligns with prior studies showing that AI is often perceived as more objective and freer from partisan bias (i.e., apolitical; Chung et al., 2024), unlike human fact-checkers (Liu et al., 2025), and hence more credible in the fact-checking process (Banas et al., 2022). On the other hand, the current discourse in research communities about AI bias and fairness highlights that AI can be too rigid and heavily reliant on unbalanced training data when fact-checking information, and therefore disproportionately benefits the majority groups in the process (Neumann & Wolczynski, 2023).

Together, these findings suggest that positive and negative machine heuristics may serve as underlying psychological mechanisms explaining users' perceptions of source bias and the credibility of fact-checking explanations, and are likely to cancel each other out, yielding a null effect in a parallel mediation model. Therefore, we propose:

H1: Automated fact-checkers will trigger the *positive machine heuristic* compared to a system that adopts human fact-checkers, which in turn will predict a) higher perceived credibility of the explanation, b) higher trust toward the system, c) lower perceived bias of the system, and d) higher reliance on the system among users.

H2: Automated fact-checkers will trigger the *negative machine heuristic* compared to a system that adopts human fact-checkers, which in turn will predict a) lower perceived credibility of the explanation, b) lower trust toward the system, c) higher perceived bias of the system, and d) lower reliance on the system among users.

How is News Fact-Checked? Comparing Two Different Approaches

Another explanation to reconcile the inconsistent results could be the different approaches that automated fact-checkers and humans usually adopt to fact-check information. The most common way for human fact-checkers to debunk misinformation is to search for past evidence that either supports or disputes the checked claim outside of the article itself (i.e., consulting the original source of the claim, going through interview transcripts; e.g., Snopes, PolitiFact), namely an evidence-based approach, which has been shown to be helpful in clearing users' misperceptions arising from health-related misinformation (Bode & Vraga, 2018). However, when it comes to automated fake news detection algorithms, information is classified primarily based on

specific features of the news presentation, such as linguistic features of the news content, the source of the news, and the contextual environment where the news is posted (e.g., repercussions and spread on social media; Reis et al., 2019). That said, it should be noted that human fact-checkers might also utilize specific features when checking information online. Scholars have proposed a set of criteria and features that human journalists might use to better discern misinformation online, including but not limited to message and linguistic features (e.g., whether the news adheres to journalistic style of reporting, the usage of hyperboles, etc.), unverified cited sources, and certain structural components on the news websites (Molina et al., 2021). In a similar vein, researchers have also developed algorithms that debunk misinformation by searching and comparing the checked claims with evidence in the database. A systematic review of fake news detection techniques reveals that while 28% of the past research articles focus on developing machine learning algorithms that use a feature-based approach, 21% of the articles focus on what they label the “knowledge-based” approach that essentially compares the checked claims with a repository that contains verified facts (Beer & Matthee, 2020). The *ClaimBuster* could be an example of an automated fact-checking system that focuses on checking evidence (Hassan, Arslan, et al., 2017) that uses machine learning, natural language processing, and database query techniques to build a collection of factual claims based on evidence from live discourses (e.g., interviews), social media, and professional news organizations to detect mismatch between the claims and the database (Hassan, Arslan, et al., 2017).

How do users make sense of why the news is flagged as false? Studies that have tested the effects of fact-checking algorithms usually do not disclose why the news is flagged as false (i.e., black-box systems; Seo et al., 2019; Zhang et al., 2021). As a result, it is still unclear how users react to automated fact-checking systems that explain why they reached a decision. Providing some information about *how* the system works could be quite beneficial to facilitate user trust in the system. Both journalists and skeptical users have expressed the need for more transparency about the process of fact-checking (Newman & Fletcher, 2017). This might be especially important for automated fact-checking systems, given that system transparency has been consistently found to be a key factor that facilitates users’ trust in AI systems (Toreini et al., 2020), reduces the perceived bias of the system (Abdollahi & Nasraoui, 2018), and enhances the likelihood of users’ reliance on the system (Schmidt et al., 2020). Therefore, we expect significantly higher trust if the system discloses the approach it uses to fact-check information (either evidence-based approach or feature-based approach), due to higher perceived transparency of the system:

H3: If the system explains the approach it adopts to fact-check information (i.e., feature-based approach or evidence-based approach), users will perceive the system to be more transparent, which will be associated with a) higher

perceived credibility of the explanation, b) higher trust toward the system, c) lower perceived bias of the system, and d) higher reliance on the system to detect misinformation online compared to a black box fact-checking system.

Transparency, however, is not our end goal. Increasingly, researchers also call for more attention to users' perception of these explanations, or human-centered explainable AI (XAI) (Miller, 2019), especially how to make lay users *understand* how (sometimes even why) AI reaches a specific decision. While transparency usually refers to the degree to which the AI system makes the information regarding itself visible to users through active disclosure, which is in direct contrast to a black box system (Turilli & Floridi, 2009), understandability is a subjective perception among users in terms of their evaluation of how much they know about the system, and is often argued to be far more important than mere transparency (Verhagen et al., 2021). Studies have shown that understanding of the system is critical for increasing users' trust and acceptance of automated systems (Ananny & Crawford, 2018) and helps users make more informed decisions with regard to their future usage of the system (Liao et al., 2020). Therefore, we propose:

H4: If the system explains the approach (i.e., feature-based approach or evidence-based approach) it adopts to fact-check information, users will perceive the system to be more understandable, which will be associated with a) higher perceived credibility of the explanation, b) higher trust toward the system, c) lower perceived bias of the system, and d) higher reliance on the system to detect misinformation online compared to a black box fact-checking system.

Exploring which approach is perceived as easier to understand and comprehend by users could be extremely valuable for developing better explainable systems, as we could focus our design effort on providing information that users truly need. While extant research mainly focuses on developing evidence-based or feature-based systems (Beer & Matthee, 2020), empirical studies do not appear to compare users' perceptions of these two approaches. Feature-based systems might be difficult for lay users to comprehend if they do not have the background knowledge in automated systems or professional journalistic practices. Given the scarcity of past empirical research, we propose the following research question.

RQ1: Will users trust evidence-based systems more than feature-based systems due to higher perceived understanding of the system?

Do Users Trust an Automated fact-checker That Interprets Evidence?

Providing factual information and evidence to bolster specific claims is usually seen as one of the uncontested key features of professional journalistic practice (Henke et al., 2020), and a unique task that only humans can perform. An interview with 24 human fact-checkers revealed that even experienced fact-checkers felt skeptical about AI in the process of claim verification (Nakov et al., 2021). Although they acknowledge the fact that automated systems might be able to perform large-scale searches of evidence in a more efficient manner, they believe that these automated systems lack the intuition and creativity to track down sources and interpret different pieces of evidence, and hence are less willing to delegate the evidence interpretation task to AI. This could be due to the fact that machines are not yet expected by fact-checkers to interpret complicated evidence. As a result, the effects of the source (human vs. machine) might be especially pronounced when the machine's performance violates a person's expectations (Lee, 2024), prompting them to rely on their prior beliefs about machines (the machine heuristic) to guide their response to AI fact-checkers. The same perception might apply to lay users as well when they interact with an automated system that adopts an evidence-based approach that highlights users' appreciation of human subjective judgment and flexibility when interpreting the evidence.

Human fact-checkers may hold this perception due to the threat of being replaced by automated systems. However, growing empirical evidence suggests that this concern might be valid. For instance, studies of evidence-based automated systems have shown that only 58% of the evidence sentences retrieved were relevant to the claim (Miranda et al., 2019), underscoring the need for human-level understanding of contextual information and subjectivity to make accurate interpretations and inferences of evidence in the fact-checking process. In this study, we seek to test whether such a perception also resonates with lay users:

H5: For fact-checking systems that use an evidence-based approach, the adoption of automated fact-checkers will more likely trigger the negative machine heuristic compared to the adoption of a human fact-checker, which in turn will be associated with a) lower perceived credibility of the explanation, b) lower trust toward the system, c) higher perceived bias of the system, and d) lower reliance on the system.

Machines are usually seen as more objective and able to analyze information at a much more granular level, hence they are more likely to be perceived as superior in analyzing a vast number of features. For instance, in the misinformation detection context, one important feature is the linguistic aspect of the information (Reis et al., 2019). Compared to human fact-checkers, it might be easier for

automated fact-checkers to pick up nuances in language in terms of the number of unique words, syllables as well as word categories (verbs, nouns, etc.) per sentence, which is a good indicator of text readability and quality (Reis et al., 2019). Furthermore, one key factor predicting users' acceptance and even higher trust of AI compared to humans in the content moderation domain is the belief that machines can make classifications based on a set of constant rules across different cases (Jhaver et al., 2018), which is more easily achieved by automated systems that adopt a feature-based approach. Compared to the evidence-based approach, the feature-based approach is less context-sensitive and is more likely to adhere to a set of pre-determined criteria (e.g., source, linguistic features, etc.; Reis et al., 2019):

H6: For an automated fact-checker, the adoption of the feature-based approach will be more likely to trigger the positive machine heuristic compared to the adoption of an evidence-based approach, which in turn predict a) higher perceived credibility of the explanation, b) higher trust toward the system, c) lower perceived bias of the system, and d) higher reliance on the system among users.

Method

To test our hypotheses and answer our research question, we conducted a 2 (Fact-checking source: Human vs. AI) $\times$ 3 (Fact-checking approach: Evidence-based vs. Feature-based vs. Black-box) between-subjects online experiment. All reported hypotheses in the manuscript were pre-registered in the Open Science Framework before data collection: <https://osf.io/qzv8w/overview>.

Participants

We recruited participants who resided in the United States from CloudResearch (originally called TurkPrime), which consists of high-quality workers from Amazon Mechanical Turk (Litman et al., 2017; $N = 291$) in July 2022. Out of a total of 312 participants, 291 passed both attention check questions embedded in the survey and constituted our final dataset. The mean age of our participants was 44.07 ($SD = 13.91$), with a range from 18 to 79. 55.7% of the participants were female ($N = 162$), 41.9% were male ($N = 122$), while 0.7% indicated their gender as other ($N = 2$), and 1.7% preferred not to provide their gender ($N = 5$). Most of our participants were Caucasian (82.1%, $N = 239$), followed by African American (7.6%, $N = 22$), and Asian (3.8%, $N = 11$). Our participants were also relatively educated with 57.4% reporting a bachelor's degree or above, and their political ideology on average was relatively neutral ($M = 4.39$, $SD = 1.87$).

We conducted an *a priori* power analysis with six conditions, two degrees of freedom, an alpha of 0.05, and a medium effect size (0.20).¹ To achieve a power of 0.8, the required sample size is 244, which is smaller than our sample of 291 (Faul et al., 2007), suggesting sufficient power for our data analyses.

Procedure

Upon providing informed consent, participants were first asked to complete a pre-stimulus questionnaire regarding control variables and their demographic information. Then, they were randomly assigned to interact with one of the six fact-checking system prototypes based on the conditions. After landing on the home page, participants first saw a welcome page that included a brief introduction of the fact-checking system called FactDeck (Figure 1(a)). In the introduction, we described how the system works, namely

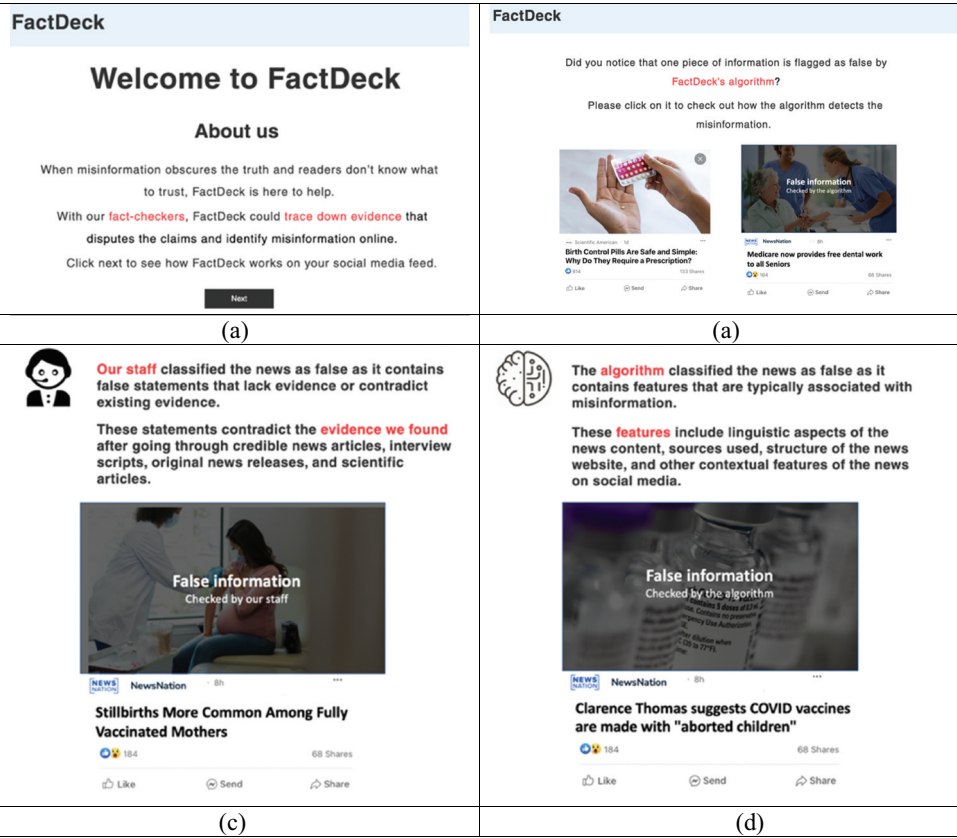


Figure 1. Screenshots of FactDeck. Figure 1a: the welcome page for FactDeck (evidence-based systems with human fact-checker); Figure 1b: the social media posts gallery page for FactDeck (systems with automated fact-checker); Figure 1c: the explanation page for FactDeck (evidence-based systems with human fact-checker); Figure 1d the explanation page for FactDeck (feature-based systems with automated fact-checker).

who checked the misinformation and the approach the system adopted. In the next page, we showed participants a gallery of six different news posts, among which one post was classified as false by FactDeck, with a one-sentence explanation that the post was flagged as false by either a human fact-checker or an automated fact-checker (Figure 1(b)), so the source cue was emphasized one more time. Finally, all participants saw a detailed explanation of why the information was flagged as false (Figure 1(c,d)). To ensure all participants finished the entire procedure, we embedded a completion code at the end of the interaction for them to input in Qualtrics to get access to the post-stimulus questionnaire, where we asked questions about the mediators and the dependent variables. All participants received \$1 compensation upon completion of the survey (median completion time of the entire study: 11.9 min).

Stimuli

We first conducted a pretest to find suitable pieces of misinformation to build our stimuli. We selected six different news story headlines, three of which scored the highest on perceived credibility and three that scored the lowest, in order to enhance the generalizability of our findings. Please refer to Supplementary Material I for more information about the pretest, as well as the final set of news headlines we used in the main experiment.

As mentioned in the procedure, we provided the manipulation of the source cues three times – on the welcome page, the news posts gallery page, and the explanation page, respectively. The manipulation for the approach cue was provided twice, once on the welcome page and the other on the explanation page (Supplementary Material II).

Measures

Except for the manipulation check questions, which followed a multiple-choice format, all items were measured on a 7-point Likert scale (1 = strongly disagree; 7 = strongly agree). Supplementary material III provides the full questionnaire.

Mediating Variables

Positive machine heuristic (PMH) and *negative machine heuristic* (NMH) were measured by adopting the scale from a previous study (Molina & Sundar, 2022). PMH was measured by two items “has machine-like precision,” and “is error free” about FactDeck (*Spearman-Brown Coefficient*² = .68, $M = 4.13$, $SD = 1.26$). NMH was measured by these two items: “is able to detect human emotion” (R), and “has human-like subjective judgements” (R)” (*Spearman-Brown Coefficient* = .66, $M = 3.99$, $SD = 1.47$).³ The two items were reverse-coded; therefore, the higher the

score on NMH, the more the system was perceived as lacking human abilities.

Perceived transparency (PT) was measured via three items (Tsai et al., 2021), e.g., “The system is transparent” and “The system shares relevant information about its underlying mechanism to users” ($\alpha = .97$, $M = 4.06$, $SD = 1.69$). *Perceived understanding* (PU) was measured with three items (Sun & Sundar, 2022), e.g., “I understood why certain information is flagged as misinformation by the system” ($\alpha = .93$, $M = 4.48$, $SD = 1.66$).

Dependent Variables

Perceived credibility of the explanation was measured via the three-item message credibility scale (Appelman & Sundar, 2016), i.e., how 1) accurate, 2) authentic, and 3) believable they found the explanations provided by the system to be ($\alpha = .95$, $M = 4.87$, $SD = 1.30$).

Trust toward the system was measured via three dimensions (Ashfaq et al., 2020; Dabholkar et al., 2009; Riquelme & Román, 2014) ($\alpha = .97$, $M = 4.33$, $SD = 1.37$), including cognitive trust (four items, e.g., “I can trust the performance of FactDeck to be good.”) ($\alpha = .98$, $M = 4.63$, $SD = 1.39$), affective trust (three items, e.g., “When it comes to misinformation detection, I can depend on the support provided by FactDeck”) ($\alpha = .90$, $M = 4.38$, $SD = 1.35$), and behavioral trust (three items, e.g., “I intend to continue using FactDeck in the future”) ($\alpha = .99$, $M = 3.86$, $SD = 1.81$).

Perceived bias of the system was measured by adopting the 5-item scale (Eveland et al., 2003), e.g., asking participants to rate FactDeck on adjectives like “biased,” “slanted,” and “objective (R)” ($\alpha = .90$, $M = 3.11$, $SD = 1.24$).

Reliance on the system was measured via three items (Khan & Idris, 2019), e.g., “My interaction with FactDeck makes me feel I can post or share facts instead of misinformation without the help of a third-party fact-checker like FactDeck.” We reverse-coded these items so that scores indicate the extent to which the participant feels that they cannot detect misinformation without the help of FactDeck ($\alpha = .85$, $M = 3.39$, $SD = 1.37$).

Control Variables

We also controlled for several individual differences in the current study, including automation bias and political ideology. *Automation bias* was controlled because it could influence users’ trust toward AI-based systems (Parasuraman & Manzey, 2010). It was measured by adopting the Automation-induced Complacency Potential Scale (Merritt et al., 2019), e.g., “When I have a lot to do, it makes sense to delegate a task to automated systems” ($\alpha = .81$, $M = 4.36$, $SD = .83$). *Political ideology* was controlled because liberal and conservative users tend to react to misinformation flagging differently (Jia et al., 2022). It was measured via one item “In politics, people

sometimes talk about liberals and conservatives. Where would you place YOURSELF on a scale from 1 to 7 where 1 means very conservative and 7 means very liberal?” ($M = 4.39$, $SD = 1.87$). In addition to the individual differences, we also statistically controlled for a contextual factor, which is participants’ *perceived newsworthiness of the news item*, to ensure that the content of the news headline does not introduce a confound in our study. It was measured via three self-developed items, namely the extent to which they thought the news article to be interesting, curious, and share-worthy ($\alpha = .93$, $M = 4.29$, $SD = 1.74$).

Data Analysis Plan

We first conducted a two-way MANCOVA (IVs: source & approach) to explore the effects of the two IVs on the mediators as proposed in H1, H2, H3, H4, RQ1, H5, and H6. To test the mediating effects proposed in H1, H2, H3, and H4, we used Model 4 of the Process macro in SPSS to run the parallel mediation models (Hayes, 2017). To test H5 and H6, we used Model 7 of the PROCESS macro, which tested the mediation relationships between approach and the DVs moderated by source cues.

Results

We report the manipulation check results in Supplementary Material IV.

Effects of Source Cues

H1 and H2 proposed that the users’ evaluations of the systems that adopt automated fact-checkers or human fact-checkers are likely to vary based on the triggering of different heuristics. Consistent with what we hypothesized, MANCOVA [$Wilks' \Lambda = .83$, $F(2, 281) = 29.64$, $p < .001$, $partial \eta^2 = .17$] revealed a main effect of source on both the PMH [Human: $M = 3.86$, $SD = 1.30$; Algorithm: $M = 4.38$, $SD = 1.17$, $F(1, 282) = 13.42$, $p < .001$, $R^2 = .05$], and the NMH [Human: $M = 3.50$, $SD = 1.37$; Algorithm: $M = 4.45$, $SD = 1.43$, $F(1, 282) = 34.02$, $p < .001$, $R^2 = .11$].

As anticipated, we did not find any total effects of source on any of our dependent variables, namely perceived credibility of the explanations [$F(1, 282) = .19$, $p = .664$, $R^2 = .00$], trust toward the system [$F(1, 282) = 1.04$, $p = .309$, $R^2 = .00$], perceived bias of the system [$F(1, 282) = .63$, $p = .429$, $R^2 = .00$], or their reliance on the system [$F(1, 282) = .03$, $p = .866$, $R^2 = .00$]. However, the relationship between the source and the DVs was mediated by PMH and NMH, such that participants tended to perceive the explanation provided by the fact-checking algorithm to be more credible (H1a supported) and trust the system more compared to human fact-checkers

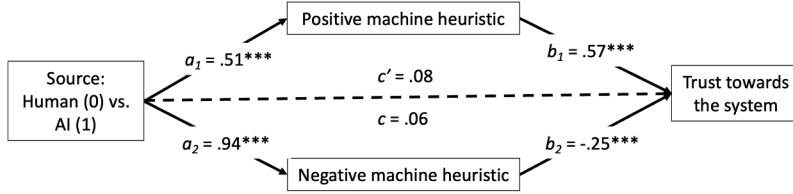


Figure 2. Effect of source cues on trust toward the system. All effects are unstandardized coefficients (b). *** $p < .001$. c' represents the direct effects of source cues on trust; c represents the total effect of source on trust.

due to the triggering of PMH (H1b supported). However, participants also tended to perceive the explanation to be more credible (H2a supported) and trusted the human fact-checker-based system more due to the triggering of NMH (H2b supported; Figure 2 and Table 1 in Supplementary Material V). The relationship between source and perceived bias of the system was only mediated by PMH (H1c supported), not NMH (H2c not supported). We did not find any indirect relationships between source and users' perceived reliance on the system, failing to support H1d and H2d.

Effects of the Misinformation Fact-Checking Approaches

H3 and H4 proposed that users will trust the black box systems due to lower perceived transparency and lower perceived understandability. MANCOVA [Wilks' $\Lambda = .63$, $F(4, 562) = 36.02$, $p < .001$, $partial \eta^2 = .20$] revealed that there is a significant difference between the three approaches on PT [$F(2, 282) = 73.45$, $p < .001$, $R^2 = .34$], and PU [$F(2, 282) = 66.48$, $p < .001$, $R^2 = .32$]. The black box systems were perceived as significantly less transparent ($M = 2.73_a$, $SD = 1.40$) compared to the other two types of systems, but there were no significant differences between feature-based systems ($M = 4.51_b$, $SD = 1.36$) and evidence-based systems ($M = 4.89_b$, $SD = 1.48$) in terms of PT. On the other hand, we found the three approaches to be significantly different from each other on perceived understandability, such that the black box systems ($M = 3.22_a$, $SD = 1.59$) were perceived as the least understandable, followed by the feature-based ($M = 4.81_b$, $SD = 1.35$) and evidence-based systems ($M = 5.34_c$, $SD = 1.27$) (Figure 3).

Furthermore, we found that systems adopting and offering explanations about their evidence-based approach and feature-based approach were perceived to offer more credible explanations (H3a supported) and were trusted more (H3b supported) than the black box systems, due to the mediation of PT. But PT did not mediate the relationship between approach cues on perceived bias of the system (H3c not supported) or reliance on the system (H3d not supported). Systems adopting and offering explanations about their evidence-

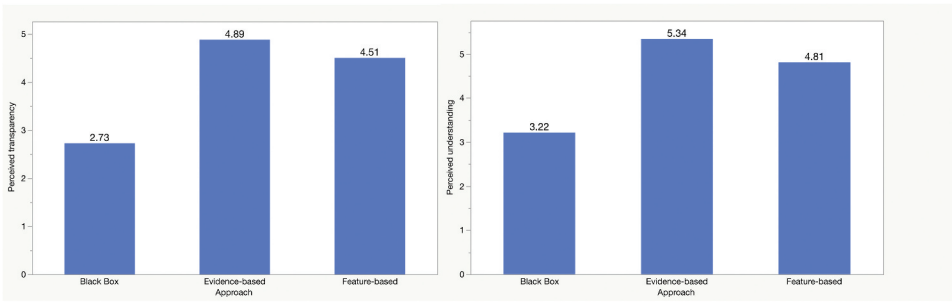


Figure 3. Effects of fact-checking approach on perceived transparency (left) and perceived understanding (right).

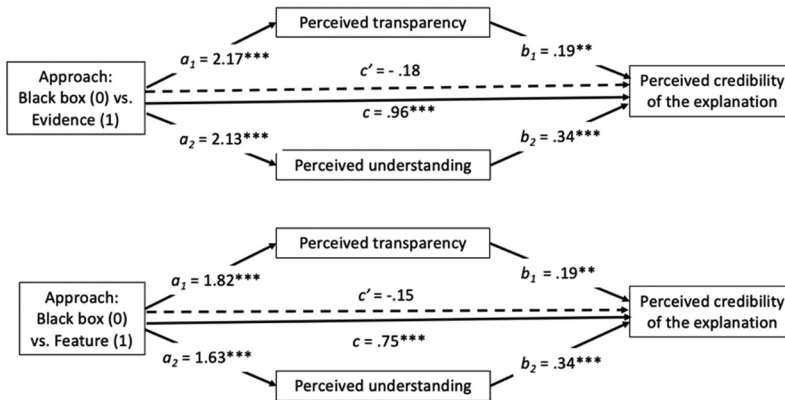


Figure 4. Effect of approach cues on perceived credibility of the explanation. Top: the comparison between black box and evidence-based approach; bottom: the comparison between black box and feature-based approach. All effects are unstandardized coefficients (b). $^{***}p < .001$. $^{**}p < .01$. c' represents the direct effects of approach cues on explanation credibility; c represents the total effect of approach cues on explanation credibility.

based approach and feature-based approach were not only perceived to offer more credible explanations (H4a supported) and trusted more (H4b supported) than the black box systems due to higher PU but were also perceived to be less biased than black box systems (H4c supported). However, we found no indirect relationship between approach cue and perceived reliance on the system through perceived understanding (H4d not supported; Table 2 in Supplementary Material V and Figure 4).

Furthermore, to answer the RQ, we found that the evidence approaches were trusted more, and perceived to be less biased; also, the explanations of evidence-based approach were perceived to be more credible compared to feature-based systems due to the mediation of higher perceived understandability (Figure 5 and Table 3 in Supplementary Material V).

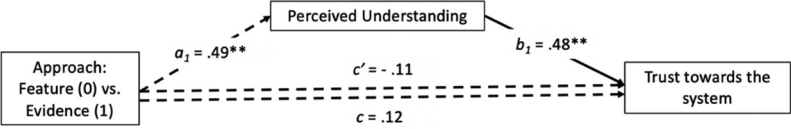


Figure 5. Effect of feature-based approach and evidence-based approach on trust toward the system via the mediation of perceived understanding. All effects are unstandardized coefficients (b). $^{**}p < .01$. c represents the total effect of approach cues on trust. c' represents the direct effects of approach cues on trust (i.e., total effect minus the indirect effect).

Interaction Effects Between Source Cues and Fact-Checking Approaches

MANCOVA [*Wilks' Λ* = .97, $F(4, 562) = 2.09$, $p = .08$, *partial η^2* = .02] revealed that there was no significant interaction effect of source and approach cues on the PMH [$F(2, 282) = .66$, $p = .520$, $R^2 = .01$] (H6 not supported), but rather a significant interaction effect on the NMH [$F(2, 282) = 3.88$, $p = .022$, $R^2 = .03$]. Specifically, when participants interacted with systems that adopted automated fact-checkers, there was no significant difference between the three approaches on NMH (though we did find a pattern such that NMH was less likely to be triggered when the algorithm-based system adopted a feature-based approach in detecting misinformation). However, with human fact-checkers, NMH was less likely to be triggered when the system adopted an evidence-based approach compared to the black box system, or when the system adopted a feature-based approach (Figure 6).

NMH was found to mediate the relationship between source and approach on users' perceived credibility of the explanations (H5a supported), trust toward the system (H5b supported) but not perceived bias of the system (H5c not supported) or perceived reliance on the system (H5d not supported; Figure 7 and Table 4 in Supplementary Material V).

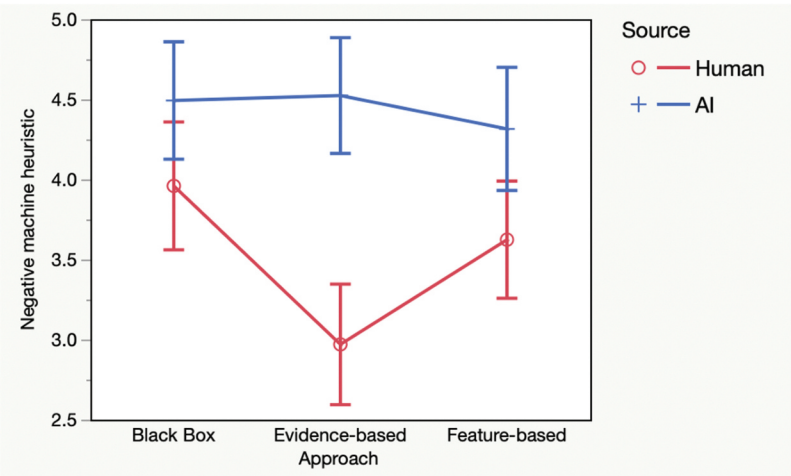


Figure 6. Interaction effects of source and approach on negative machine heuristic.

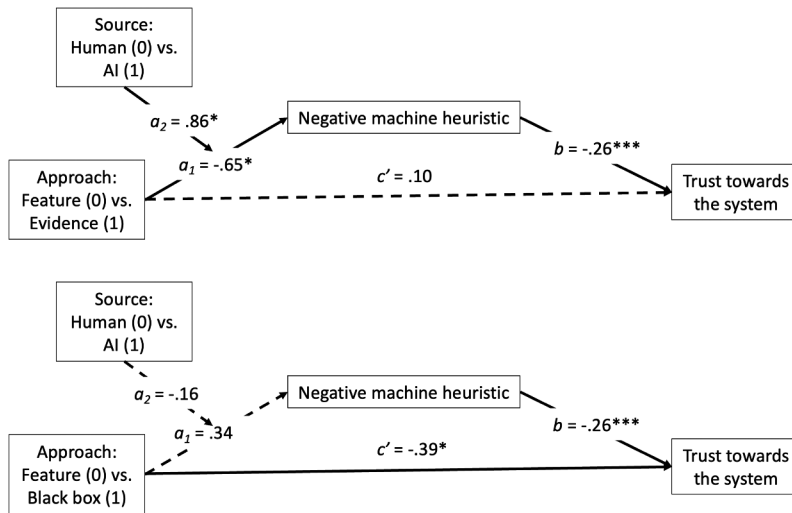


Figure 7. Effect of approach and source cues on trust toward the system. Top: the comparison between feature-based approach and evidence-based approach; bottom: the comparison between feature-based approach and black box system. All effects are unstandardized coefficients (b). $^{***}p < .001$. $^*p < .05$. c' represents the direct effects of approach cues on trust toward the system.

Summary of Results

We found that the relationship between the source cues and users' evaluation of the system is mediated by both the positive machine heuristic and the negative machine heuristic, which worked in opposing directions (canceling out each other), resulting in a non-significant total effect. We also found that participants tend to trust the systems more if any explanation is provided, compared to a black box system, due to higher perceived transparency and higher perceived understanding of the system. Furthermore, explanations based on evidence were perceived to be far easier to understand compared to a feature-based approach. Finally, we found an interaction effect between source cues and approach cues such that when the system employed human fact-checkers who adopted the evidence-based approach, there was a lower likelihood of negative machine heuristic being triggered, thus predicting higher trust in the system.

Discussion

By examining the underlying psychological mechanisms that explain why users trust automated versus human fact-checkers, our study offers important theoretical implications for both the fact-checking and human–AI interaction literature, which we detail below.

Theoretical Implications

First of all, our discovery of two distinct mediators rather than an overall total effect of fact-checking sources on users' evaluation of the fact-checking system helps explain the mixed findings in the literature regarding users' trust toward AI systems that classify misinformation (Jia et al., 2022; Seo et al., 2019; Yaqub et al., 2020; Zhang et al., 2021). The extent to which the positive or the negative machine heuristic is triggered seems to be the key to understanding why users sometimes trust and sometimes distrust AI systems. Theoretically, these findings corroborate the HAI-TIME model, which proposes that users' formation of trust and perceptions toward AI are dictated by the nature of the cognitive heuristics triggered by cues on the AI interface (Sundar, 2020). More importantly, we found that both positive and negative machine heuristics are critical in the context of users' interaction with fact-checking systems. This result not only empirically supports the theoretical distinction between positive and negative machine heuristic (Molina & Sundar, 2022) but also encourages future research on human-AI interaction to examine not only the positive aspects of machines compared to humans (i.e., the positive machine heuristic) but also the negative aspects (i.e., the negative machine heuristic), rather than focusing primarily on the total effects of source (AI vs. human) on users' trust and evaluation of emerging AI systems. Understanding these underlying psychological mechanisms helps explain *why* users form trust or distrust in AI.

It is also important to examine the boundary conditions under which negative or positive machine heuristics are more likely to be triggered. In general, our results reveal that users' distrust toward AI in the misinformation detection context largely lies in their concern that the algorithm lacks human subjective judgment and is unable to detect human emotions. More importantly, we found that this might especially be the case when the system uses an evidence-based approach to fact-check information. In line with our hypothesis, we found that users trust humans more in finding evidence compared to AI, probably because they are reminded about machines' weaknesses when the fact-checking system adopts an evidence-based approach (i.e., triggering of negative machine heuristic). Humans are seen as more likely to detect nuances in human language, which is especially valued by users in the process of interpreting evidence. This finding corroborates a previous study, which found that user trust in machines is attributed to their mechanical abilities, whereas their trust in humans is attributed to their intellectual abilities relating to subjective decision-making (Lee, 2018). By directly capturing both negative and positive machine heuristics, our study not only unveils the psychological mechanisms underlying why machines are distrusted for tasks requiring human-level abilities but also identifies conditions under which a specific cognitive heuristic (in this case, the negative machine heuristic) is more likely to be triggered (i.e., when AI is used to find and corroborate evidence). Prior

research has generally shown that merely disclosing the source (i.e., AI vs. human) can trigger both positive and negative machine heuristics (Molina & Sundar, 2022). Few studies, however, have examined the conditions under which one heuristic is more likely to be activated than the other. We hope our study encourages future research to systematically investigate the boundary conditions that determine the triggering of positive or negative machine heuristics.

Aside from the source of fact-checking, our data show that the approach used for fact-checking is also perceptually consequential. In general, our results corroborated previous studies regarding the importance of AI transparency in users' trust, since we found that the provision of any explanation of how the system fact-checked information is appreciated by users, as it makes the system appear more transparent and understandable (Toreini et al., 2020). Furthermore, we found that explanations about the evidence-based approach were perceived as more understandable compared to feature-based explanations even though both are perceived as equally transparent. This finding might also help explain the contradictory effects of previous research about trust toward AI in the fact-checking context (Toreini et al., 2020). How algorithms fact-check information is usually not disclosed to users, leading them to rely on their folk theories or mental models to make inferences about how the system operates (Toreini et al., 2020). As a result, it is highly likely that users think that the algorithm is using some preset rules to classify misinformation, namely a feature-based approach. The higher perceived understanding of evidence-based approaches could be explained by users' familiarity with this approach, as it is widely adopted across fact-checking platforms (e.g., Snopes), and their expectations of seeing evidence to bolster claims in the news context (Henke et al., 2020).

Our finding also stresses the importance of having human-in-the-loop AI fact-checking systems that could involve human intelligence to understand complex human language (Demartini et al., 2020). The fact that we found no total effect of the source on any of our dependent variables highlights the importance of facilitating the collaboration of human fact-checkers and AI so we could leverage the power of both machines and humans in the fact-checking context. In other words, AI can help human fact-checkers to analyze, search and compare information at larger scale and at a faster pace (Nakov et al., 2021), and humans can help discern the nuances that are unlikely to be detected by AI (Sundar, 2020). Human-AI collaboration is found to be both helpful in improving the performance (Guo et al., 2019) and in the explainability of AI misinformation detection models (Nguyen et al., 2017). Future studies could further test if the involvement of humans, or AI-assisted human fact-checking, could be even more likely to be trusted by users, by maximizing the potential for triggering the positive machine heuristic and minimizing the invocation of the negative machine heuristic.

Practical Implications

In addition to its theoretical implications, our study offers several practical implications for fact-checking practitioners.

We found that both positive and negative machine heuristics are critical in the context of users' interaction with fact-checking systems, providing guidance to practitioners who would like to increase users' trust and enhance their interaction experience with fact-checking algorithms. Our results suggest possibilities for us to shape user experience by embedding simple cues on the interface that reveal who has done the fact-checking and how information was fact-checked. For instance, since the mediating effect of the positive machine heuristic suggests that users' trust in machines can be increased through an emphasis on the positive aspects of machines, an interface design that prominently positions AI as the source, and touts such positive aspects of machines as accuracy and scalability, can enhance the user experience of AI fact-checkers.

But more importantly, as revealed in our study, fact-checking is widely seen as a subjective task by lay users. Therefore, focusing on the effort invested in training the algorithms to take into account the contextual background and subjectivity of the information might be the key to decreasing the possibility of triggering the negative machine heuristic, which is known to impair users' trust toward automated fact-checkers. For instance, NLP (Natural Language Processing) researchers have started to implement contextualized rather than static word embeddings in language models to classify misinformation (Samadi et al., 2021). Contextualized word embeddings create various word representations for a target word based on different language contexts, so a target word can have multiple representations or vectors based on how it was used in the text (Suárez et al., 2020). Although contextualized word embeddings might be more beneficial in improving system performance in classifying short news statements than classifying long news articles (Samadi et al., 2021), and in the process of increasing its ability to detect subjectivity, it could also show human-like social and racial biases which researchers have also started to address (Guo & Caliskan, 2021). Communication of such technical advances in detecting linguistic nuances to users can serve to inhibit the invocation of the negative machine heuristic. Even documenting these kinds of effort put into promoting AI's ability to detect subjectivity in human language on the interface might be helpful in alleviating user concerns and dispelling their negative preconceptions of AI's limitations.

At a broader level, the invocation of both positive and negative machine heuristics when users interact with an AI system suggests rich design possibilities for embedding cues on the interface that can enhance user experience. Specifically, we could design interfaces with cues that focus on different aspects, styles, and features of decision-making by the underlying algorithm

to trigger either a positive or negative machine heuristic as part of an effort to help users calibrate their trust toward an automated system. This could include emphasizing the precision and objectivity that an AI system can bring to information classification, in keeping with our finding that when users think about such positive attributes of machines, they are likely to trust the AI's judgment. Notably, the effect of source on perceived bias was only mediated by the positive machine heuristic (Table 3 in Supplementary Material V), suggesting that machines are assumed to be bias-free, but this could well be a false perception considering that many AI algorithms are trained on biased data. In fact, the issue of fairness of algorithms has become a growing concern among AI researchers, not only in the fact-checking context (Park et al., 2022), but in several other contexts as well (Bellamy et al., 2018). The good news is that researchers in academia and the industry are both dedicating considerable research effort toward reducing bias and improving fairness of AI systems (Jhaver et al., 2018). To better calibrate users' perceptions, expectations, and trust (both distrust and unwarranted trust) toward AI classification, we should both inform users about the presence of potential biases in the algorithm and share with them the effectiveness of measures put in place to mitigate them.

Finally, our findings also stress the importance of creating a more cognitively accessible explanation if the fact-checking system relies on features to classify misinformation in both human and automated fact-checkers. In the current study, we offered global-level explanations which might be inherently abstract and less easy to understand for laypersons. One potential solution is to provide more visual cues for users to better comprehend how machines make decisions (e.g., highlight, heatmap, etc.; Lai & Tan, 2019). Another possibility is to embed interactivity affordances on the interface, so that users engage in more systematic information processing of the explanation by interacting with the system (i.e., the action route of the HAI TIME model; Sundar, 2020). For instance, a recent study has shown that interactive explanations that enable users to interact with each feature that AI adopts and to observe changes in AI's predictions (i.e., counterfactual examples) help enhance their understanding of the system (Liu et al., 2021). Future studies could test various forms of explanations (local and global), and ways of presenting the explanation to increase users' understanding of feature-based explanations.

Limitations

Our study is not free from limitations. The interaction between our participants and the fact-checking system was rather brief and one-shot, so what we observe in the current study might not hold true when using the system for a prolonged period of time (Yin et al., 2019). In order to maintain internal validity, we needed to make the content constant across conditions. Future studies could conduct field experiments to test the effects we observed in the current study.

Furthermore, we did not measure users' objective reliance, but only perceived reliance. Future studies could test whether the increased perceived reliance is, in fact, predictive of their actual reliance on the system as well as their independent ability to detect misinformation online without the help of the system. Additionally, we measured perceived transparency and understanding right before the measures of the dependent variables in one setting, which, strictly speaking, only allows for correlational evidence for the mediators and dependent variables (Chan et al., 2022). Future studies could replicate our findings by varying the degree to which the fact-checking system is transparent and understandable by lay users to test the causal relationship between the mediators and the dependent variables. Furthermore, following Bellur and Sundar (2014), the best way to operationalize and test the triggering of cognitive heuristics is to capture both the construct accessibility of the heuristic (i.e., belief in positive machine heuristic and negative heuristic), and the self-reported use of the heuristic when encountering cues that could trigger this heuristic (i.e., positive and negative machine heuristic perception as mediators), and see if there is a moderated mediation relationship: the cue's effects on mediators are moderated by users' prior belief in the cognitive heuristic. Future studies could replicate our findings by capturing both constructs. We would also like to point out that there could be alternative mediators that explain the observed effects, which future studies should investigate. Our sample size was relatively small and was intended to detect medium effect sizes. As a result, our statistical power to detect smaller effects or to test more complex mediation models was limited. Future studies should involve larger sample sizes to improve statistical power and replicate our findings in more diverse samples and non-Western countries. In addition, more field experiments are needed, where demand effects (such as expressing higher trust in fact-checking results) are less pronounced than in controlled laboratory environments.

Conclusion

Understanding users' perceptions and trust in automated fact-checkers has become critical as algorithms are being widely deployed by various online platforms to curb the spread of misinformation. We found in the current study that users' trust in the system depended largely on whether the interface triggered the positive machine heuristic (algorithm is more objective and precise than human), or the negative machine heuristic (algorithm lacks human subjective judgment). The negative machine heuristic was also more likely to be triggered when an algorithm was implemented to interpret evidence rather than detect telltale features. Our study extends current literature regarding the psychological mechanisms underlying users' formation of trust

toward AI, and suggests exciting design possibilities for triggering appropriate cognitive heuristics in the minds of users and earn their trust in a socially responsible manner.

Notes

1. We chose a medium effect size (Cohen's $f = 0.20$) based on previous literature investigating users' perceptions of and trust in fact-checking labels, which generally reported medium effect sizes when effect size statistics were provided (e.g., Cohen's $d = 0.21$ in Zhang et al. (2021)). We also considered literature comparing users' evaluations of human versus machine sources in the news context (see the meta-analysis by Huang and Wang (2023), of 30 studies, Cohen's $d = -.15$, which falls between a small and medium effect size). However, this decision may limit our statistical power to detect smaller effects.
2. Recommended by (Eisinga et al., 2013) as the preferred indicator of reliability for a two-item scale while Cronbach's alpha is more appropriate for scales with 3 or more items.
3. Originally in our pre-registration, we proposed an additional hypothesis regarding a third mediator for the effects of the source, namely the authority heuristic, or the perception that the source is an authority in fact-checking and hence was more trusted by users. However, we observed that the authority heuristic was highly correlated with the positive machine heuristic ($r = .50, p < .01$) and the negative machine heuristic ($r = -.44, p < .01$), while the correlation between the positive machine heuristic and the negative machine heuristic was $r = -.13, p < .05$. To avoid reporting redundant results (the mediating roles of negative machine heuristic and authority heuristic regarding the effects of source on our dependent variables were highly similar), we decided to exclude the authority heuristic from the analyses presented here.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

Author contributions

CRediT: **Mengqi Liao:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Validation, Visualization, Writing – original draft, Writing – review & editing; **Sian Lee:** Conceptualization, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing; **Annie Dooley:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – original draft, Writing – review & editing; **Aiping Xiong:** Conceptualization, Investigation, Methodology, Supervision, Writing – review & editing; **S. Shyam Sundar:** Conceptualization, Investigation, Methodology, Project administration, Supervision, Writing – original draft, Writing – review & editing.

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Data Availability Statement

The data underlying this article will be shared on reasonable request to the corresponding author.

Open Scholarship



This article has earned the Center for Open Science badge for Preregistered. The materials are openly accessible at <https://osf.io/qzv8w/overview>.

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